Git and GitHub

Beginner Study Notes and Practice Guide

This short guide introduces core concepts and provides practice ideas. It is an original learning summary, not a substitute for a complete textbook.

Git and GitHub

Git tracks changes to files locally. GitHub hosts Git repositories online and supports collaboration.

Repository basics

A repository stores project files and their history. A commit records a snapshot of changes.

Common workflow

Use git status to inspect changes, git add to stage files, and git commit to save a snapshot.

Remote repositories

A remote such as origin points to a hosted repository. git push uploads local commits; git pull fetches and integrates remote changes.

Branches

Branches let you develop work separately before merging it into another branch.

Collaboration

Pull requests propose changes for review. Issues help track bugs, tasks, and ideas.

Practice

Create a repository, commit a README, create a branch, make a change, and merge it.

Practice checklist

- Review the concepts and write a small example.
- Run your example and test an edge case.
- Explain the solution in your own words.